

2	a	Simple harmonic motion is an oscillatory motion in which the <u>acceleration</u> is <u>proportional</u> to the <u>displacement from a fixed point</u>, and acceleration is always directed towards that point. (or acceleration is in the opposite direction to the displacement).	B1 B1
	b(i)1.	$T = 0.300 \text{ s}$ either $\omega = 2\pi / T$ or $\omega = 2\pi f$ and $f = 1 / T$ $\omega = 2\pi / 0.300$ $= 20.944$ $= 20.9 \text{ rad s}^{-1}$ (accept 2 s.f)	C1 A1

